



Deep Learning and Applications Lab1

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Outline



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A. Environment Setup

A-1. Dual boot system

A-2. WSL2

A. Setup D415 Camera and Record Data

[LAB1 README](#)



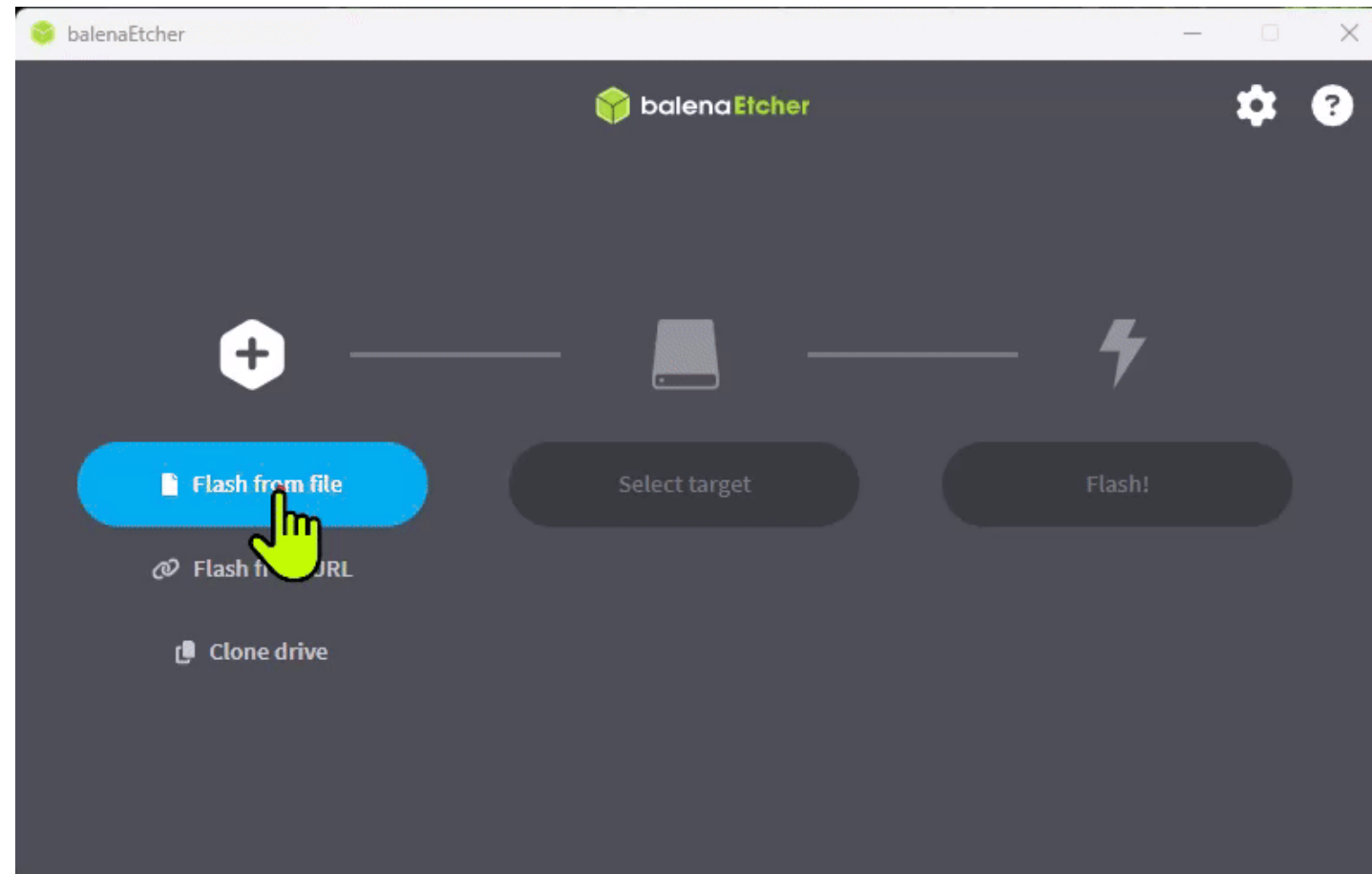
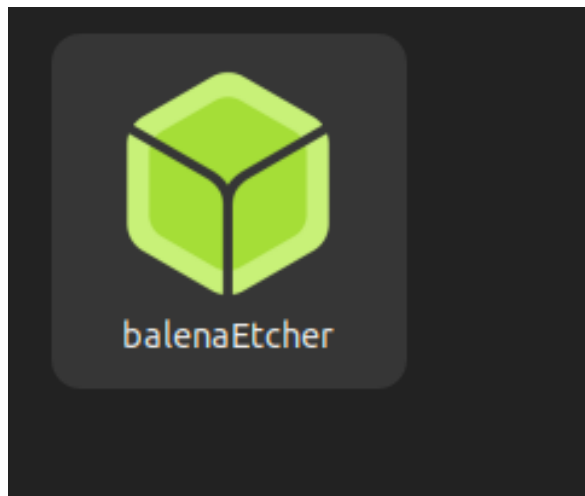
Dual boot system

- Your computer equipped with an **NVIDIA GPU**
- Disk Partitioning
 - To give Ubuntu its own system area
 - To protect your existing Windows files
- Recommended Partition Size at least 100GB
- If you don't plan to use a dual-boot system in the future, it's recommended to choose another method —otherwise, you'll need to merge the previously partitioned disk space later.



Create a bootable Ubuntu USB

- Empty Bootable USB drive *1 (16GB)
- Download Ubuntu ISO: <https://ubuntu.com/download/desktop>

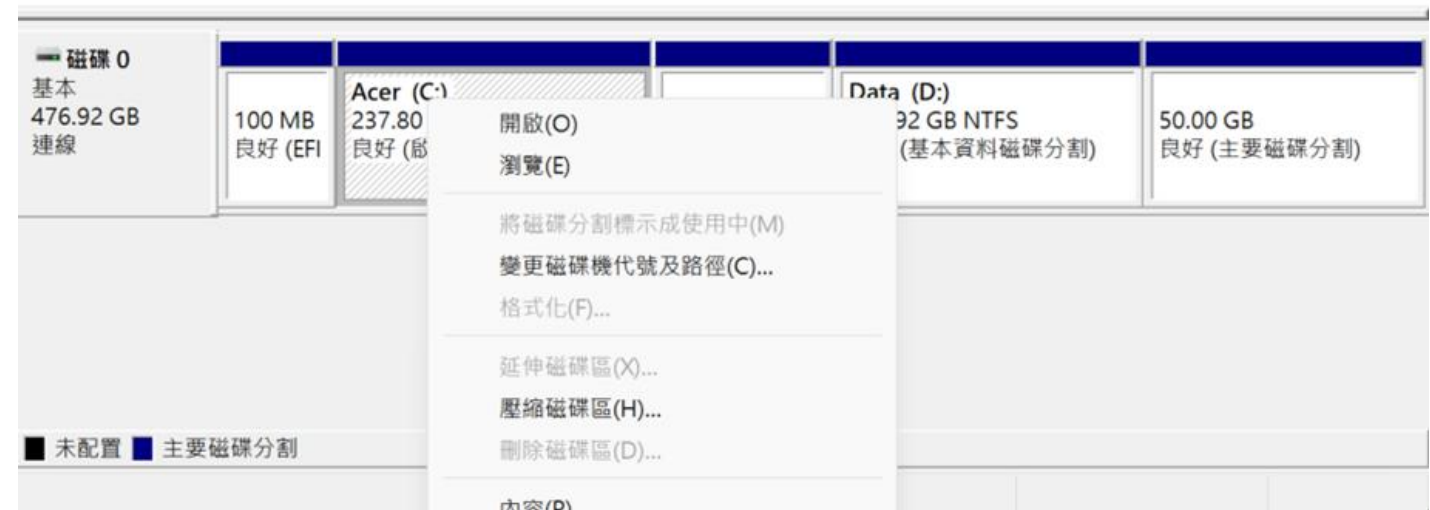


Prepare in windows

- Press **win+R**, type Diskmgmt.msc
- Select shrink volume

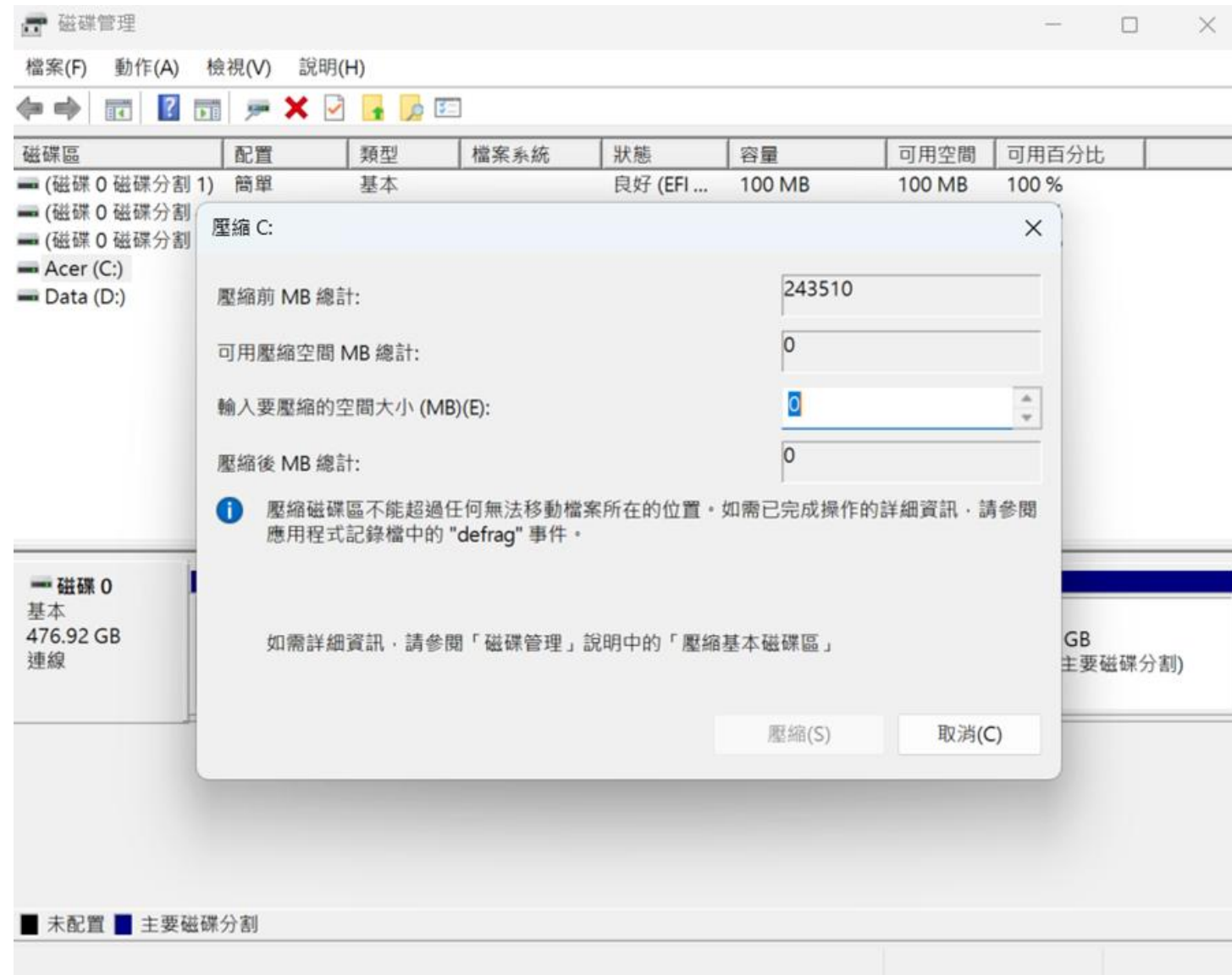


磁碟區	配置	類型	檔案系統	狀態	容量	可用空間	可用百分比
(磁碟 0 磁碟分割 1)	簡單	基本		良好 (EFI ...)	100 MB	100 MB	100 %
(磁碟 0 磁碟分割 4)	簡單	基本		良好 (修復...)	1.10 GB	1.10 GB	100 %
(磁碟 0 磁碟分割 6)	簡單	基本		良好 (主要...)	50.00 GB	50.00 GB	100 %
Acer (C:)	簡單	基本	NTFS	良好 (啟動...)	237.80 GB	17.05 GB	7 %
Data (D:)	簡單	基本	NTFS	良好 (基本...)	187.92 GB	56.70 GB	30 %



Prepare in windows

- 1GB=1024MB



Enter Bios



- Enter your Bios(ASUS:F2)
- disable the **Security boot**
- Enter boot menu
- Move your USB drive to the top of the Boot priority list
- You can watch a tutorial [Video](#)

A-2 WSL2



- If you do not have enough storage space for a dual-boot system
- Allows running Linux directly on Windows without a virtual machine
- Provides a full Linux environment for developers
- Integrates Windows and Linux file systems
- Supports Docker, and ROS 2 development
- Ideal for cross-platform software and robotics projects

WSL2



- [WSL Install](#)

```
$ wsl --install
```

- Verify your WSL version:

```
$ wsl --version # Ensure that your version is at least 2.0
```

- **Restart your computer** when prompted

Install WSL command

You can now install everything you need to run WSL with a single command. Open PowerShell in **administrator** mode by right-clicking and selecting "Run as administrator", enter the `wsl --install` command, then restart your machine.

PowerShell

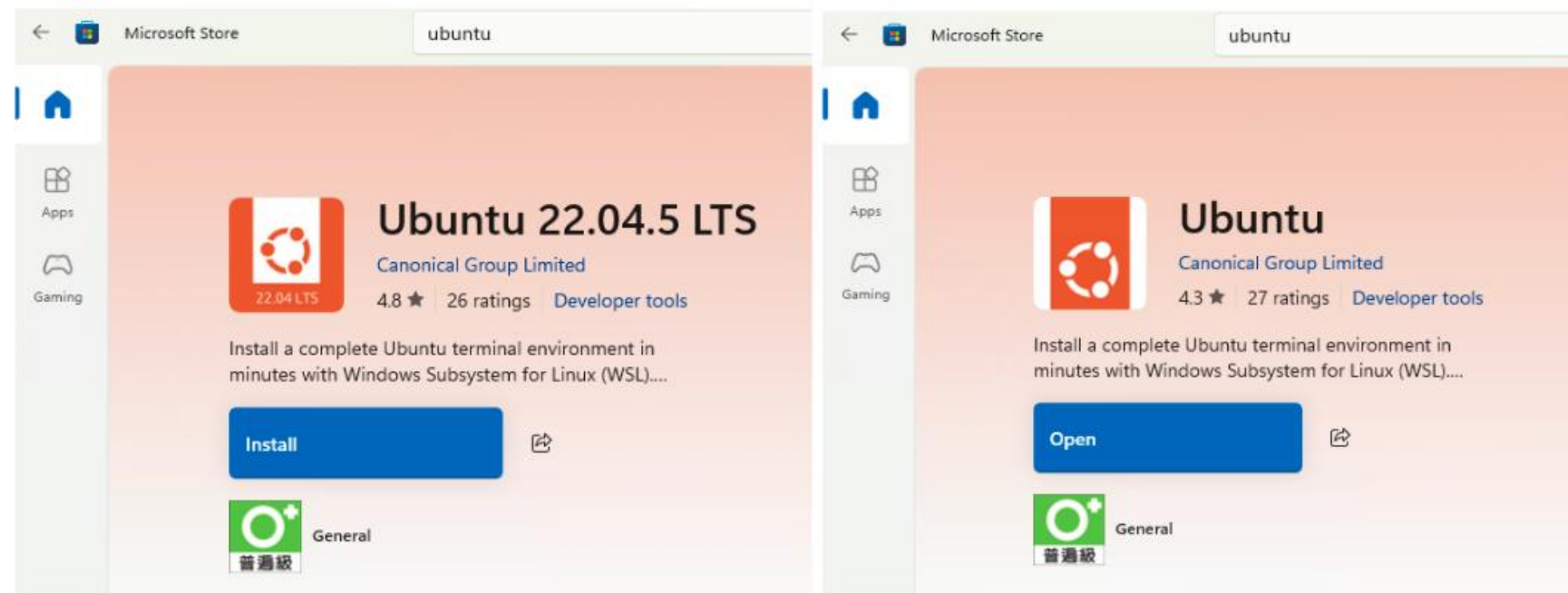
 Copy

```
wsl --install
```



Install Ubuntu for WSL

- Open the Microsoft Store, install Ubuntu 22.04 LTS (or Ubuntu)
- Launch Ubuntu, and set your username and password when prompted



```
Installing, this may take a few minutes...  
Please create a default UNIX user account. The username does not need to match your Windows username.  
For more information visit: https://aka.ms/wslusers  
Enter new UNIX username: |
```



Setup Ubuntu

- Update the system:

```
$ sudo apt update && sudo apt upgrade -y
```

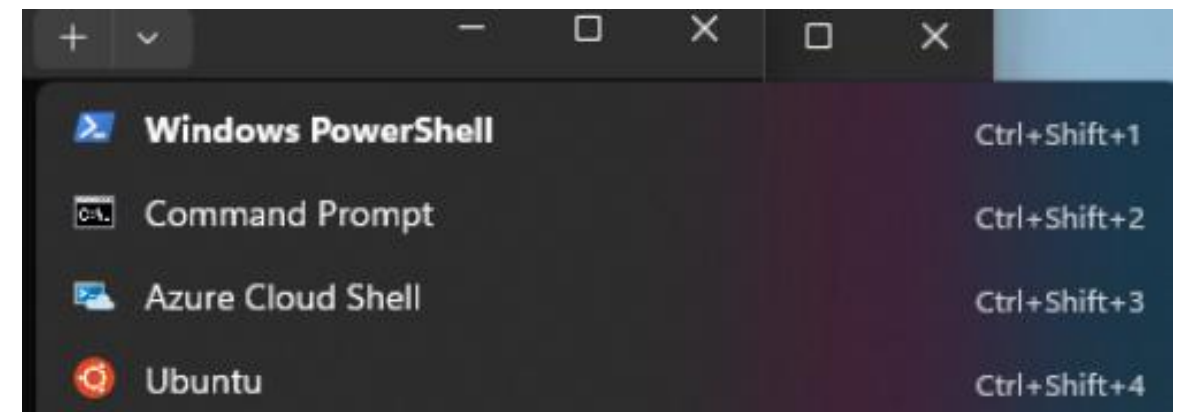
- The next time you want to open WSL:

1. Type Ubuntu in the Windows search bar
2. Open PowerShell and type:

```
$ ubuntu
```

- Open Visual Studio Code inside WSL:

```
$ code .          # to start VScode
```





Install Docker-Desktop

- Download Docker Desktop for Windows from the official site:
([Docker Desktop](#))
- You can confirm your system type in Device Manager → System Information (If it shows *x64-based processor*, select AMD64.)

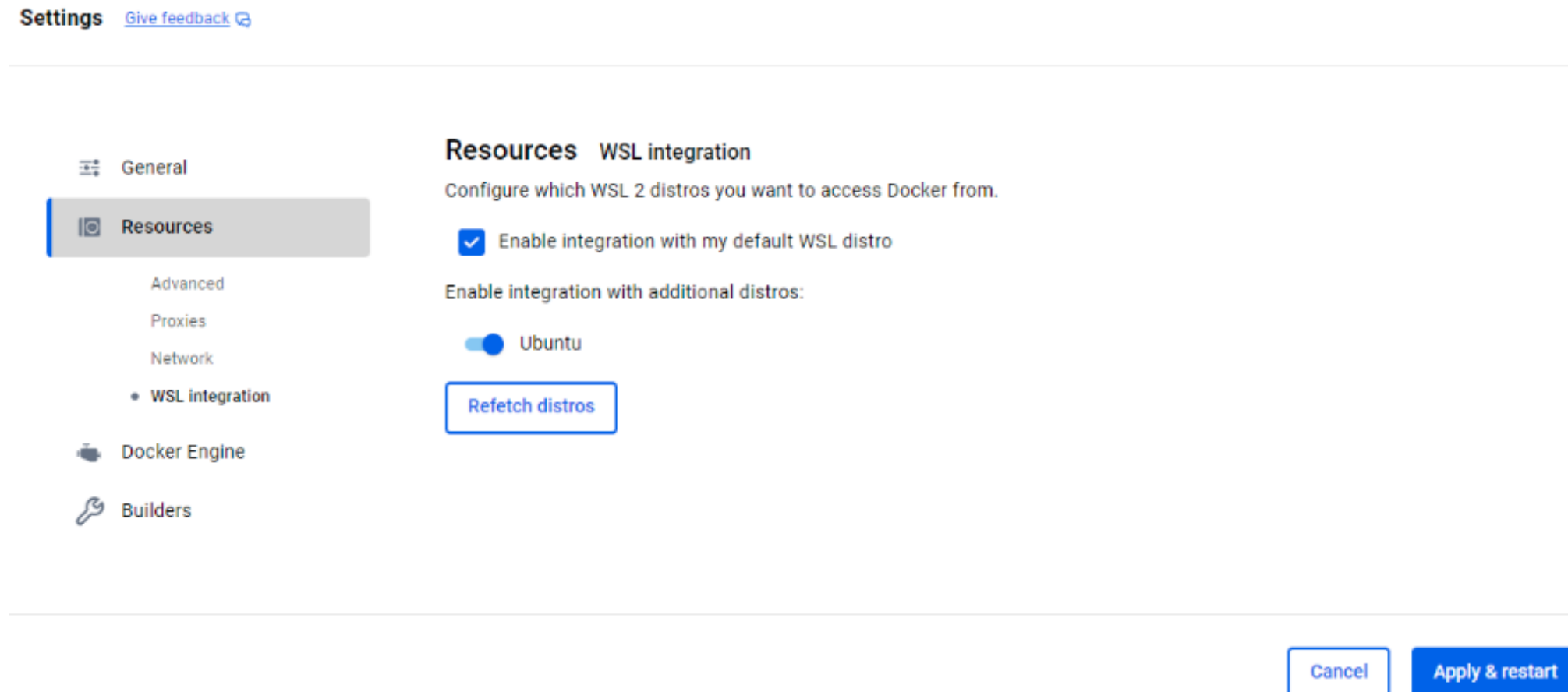


Docker-Desktop Setting



- Go to Settings → Resources → WSL Integration, enable “Enable integration with my default WSL distro” and ensure Ubuntu is selected.

Then click Apply & Restart.





Connect the Camera to WSL

- This step allows your WSL environment to access a USB camera
- Open PowerShell as **Administrator** and run

```
$ winget install --interactive --exact dorssel.usbipd-win
```

- Find the camera, list all connected USB devices:

```
$ usbipd list # find the camera BUSID
```

- Bind the camera:

Example:

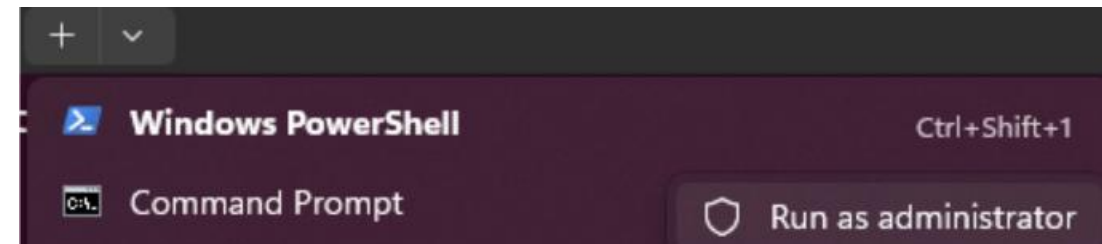
```
$ usbipd bind --busid 1-25
```

- Attach the camera to WSL:

```
$ usbipd attach --wsl --busid 1-25 --auto-attach
```

- Verify that the camera is recognized inside Ubuntu (**wsl terminal**)

```
$ lsusb
```



```
PS C:\Users\HRCla> usbipd bind --busid 1-25
PS C:\Users\HRCla> usbipd list
Connected:
BUSID  VID:PID  DEVICE                                STATE
----  -
1-11   1462:7d98  USB 輸入裝置                          Not shared
1-14   8087:0033  Intel(R) Wireless Bluetooth(R)        Not shared
1-25   8086:0ad3  Intel(R) RealSense(TM) Depth Camera 415 Depth, Intel(R) ... Shared
```

```
hrcLab@HRClab-Lester:~$ lsusb
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 002 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
Bus 002 Device 006: ID 8086:0ad3 Intel Corp. RealSense D415
```



B. Setup the D415 and record data

DL APP Repo

- **Clone the Repository**

```
git clone --recursive git@github.com:hrc-pme/Deep\_learning\_d415\_ROS2.git
```

- **Two Type of Version**

1. GPU Version

- a. PyTorch Version

- b. `./run_gpu.sh`

2. CPU Version

- a. If you are using VMware, you only can use CPU version.

- b. `./run_cpu.sh`

- **Build ROS 2 Workspace**

1. Remind to colcon build

```
cd ~/Workspace/ros2_ws
```

```
colcon build --symlink-install
```

```
# Or simply use `cb` instead of typing the full command
```

B. Setup the D415 and record data

RealSense D415 Camera Setup

1. Find Camera Serial Number

```
rs-enumerate-devices -s
```

Expected Output

Device Name	Serial Number	Firmware
Intel RealSense D415	241222061237	5.16.0

2. Configure Camera Serial Number

1. Open the launch file:

```
~/home/hrc/Workspace/ros2_ws/src/realsense-ros/real  
  
# You can change multi setting here
```

2. Update the serial number:

```
{'name': 'serial_no', 'default': "'YOUR_CAMERA_SERIAL'"}
```

Recording ROS Bags

1. Configure Recording Settings

Edit the configuration file:

```
~/Workspace/ros2_ws/src/bag_recorder/config/bag_recorder.yaml
```

Default configuration:

```
topics:  
  - /tf_static  
  - /camera/camera/color/image_raw/compressed  
  
output_dir: ~/Workspace/Outputs/rosbags/test
```

2. Start Recording

We will start to record a rosbag with camera topics.

Choose one of these methods:

Using GUI will be easier to record multiple bags.

Option 1: Using GUI

```
ros2 launch bag_recorder record_with_gui.launch.py
```

Convert ROSBags to MP4 Video

Edit the configuration file:

```
~/Workspace/Script/config/bags2mp4.yaml
```

Default configuration:

```
jobs:  
  - input_paths: # Paths to bag files  
    - "/home/hrc/Workspace/Outputs/rosbags/test"  
    output_root: "/home/hrc/Workspace/Outputs/mp4/test"  
    topic: "/camera/camera/color/image_raw/compressed" # Image topic. If omit → auto-pick if only one; error if multiple
```

NOTES: If Image topic omit → auto-pick if only one; error if multiple

Convert ROSBags to MP4 Video

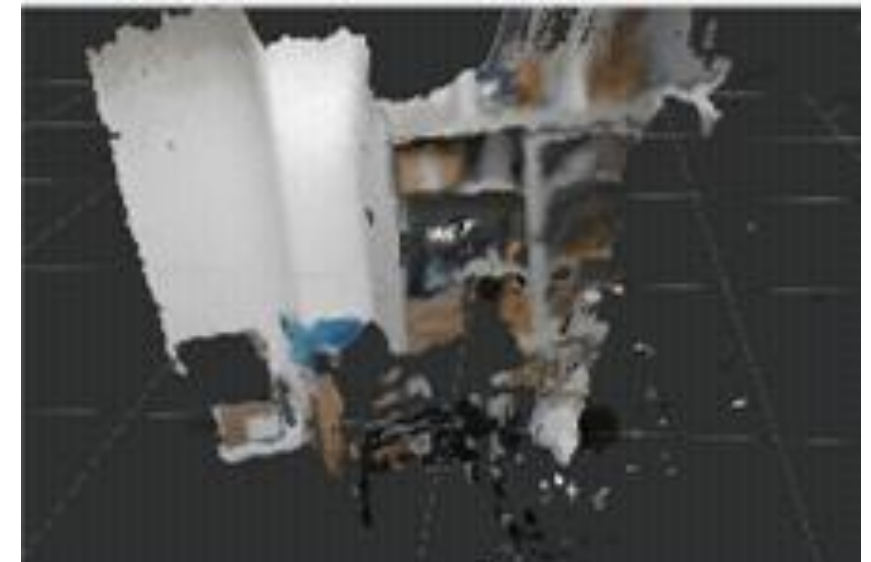
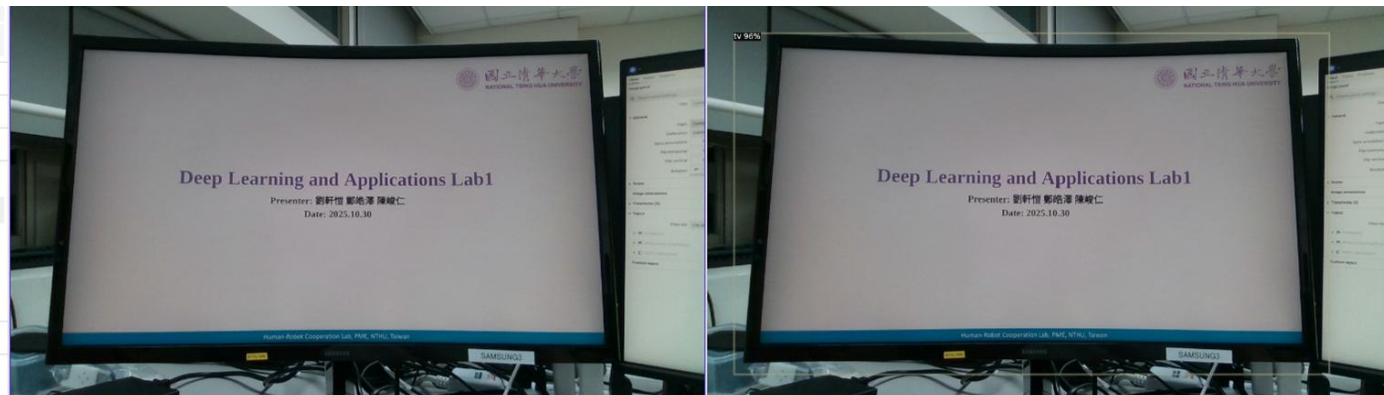
```
cd ~/Workspace/Script  
python3 bag2mp4.py
```

[DL APP Repo](#)



Access Image/Point Clouds via ROS

- Launch the camera node
- Visualize images using RViz2 or Foxglove Studio
- Confirm the point cloud topic (**Remind to open the points**)
- Visualize Point Clouds with RViz2 → Add → PointCloud2 → select topic





WSL Common Error

- Error **0x004000d** (WSL Not Enabled)
 - Windows Subsystem for Linux feature is not enabled
 - Required Windows features are not turned on
 - Virtualization technology (VT-x/AMD-V) is disabled in BIOS
- Error **0x800701bc** (Kernel Version Outdated)
 - WSL2 requires an updated Linux kernel for proper operation

[solve problem](#)